

Web Development Interview Questions & Answers (HTML, CSS, JavaScript)

Section 1: HTML (25 Questions)

Q: What is HTML? Why is it important in web development?

A: HTML defines the structure of web pages using elements like headings, paragraphs, links, and forms.

Q: Difference between HTML and XHTML?

A: XHTML is stricter, requiring closed tags, lowercase, and quoted attributes.

Q: What is semantic HTML? Give examples.

A: Semantic HTML uses meaningful tags like `<header>`, `<nav>`, `<article>`, improving SEO and accessibility.

Q: Difference between inline and block-level elements?

A: Inline elements don't start on a new line (`<span>`), block elements do (`<div>`, `<p>`).

Q: What is the difference between `<head>` and `<header>` tags?

A: `<head>` holds metadata like title/CSS, `<header>` is visible page header.

Q: What are void elements in HTML?

A: Void elements don't need closing tags, e.g., `<img>`, `<br>`, `<hr>`.

Q: Difference between `<strong>` vs `<b>` and `<em>` vs `<i>`?

A: `<strong>/<em>` are semantic, `<b>/<i>` are purely stylistic.

Q: What is the role of the alt attribute in images?

A: Provides alternative text for accessibility and SEO.

Q: Difference between `<section>`, `<div>`, and `<article>`?

A: `<section>` groups related content, `<article>` is independent, `<div>` is generic.

Q: What is the difference between absolute, relative, and fixed URLs?

A: Relative depends on current path, absolute includes full domain, root-relative starts at site root.

Q: Difference between `<link>` and `<script>` tags?

A: `<link>` includes CSS, `<script>` includes JavaScript.

Q: What is the difference between `async` and `defer` in `<script>` loading?

A: `async` runs immediately, `defer` runs after HTML parsing.

Q: What is the difference between cookies, localStorage, and sessionStorage?

A: Cookies are sent with requests, localStorage persists, sessionStorage clears on tab close.

Q: What is the difference between `<canvas>` and `<svg>`?

A: `<canvas>` is pixel-based dynamic graphics, `<svg>` is scalable vector graphics.

Q: Difference between GET and POST methods in forms?

A: GET sends via URL, POST sends via body and is more secure.

Q: Difference between id and class attributes?

A: id is unique, class is reusable across elements.

Q: Difference between required, disabled, and readonly attributes?

A: required must be filled, disabled is uneditable, readonly visible but not editable.

Q: Difference between <iframe> and <embed>?

A: <iframe> embeds webpages, <embed> embeds media.

Q: What is the difference between HTML entities and characters?

A: Entities like ensure special characters render safely.

Q: Why is <!DOCTYPE html> important?

A: Declares HTML5 standard, avoids quirks mode.

Q: Difference between inline SVG and ?

A: Inline allows CSS/JS styling, is static.

Q: What is the difference between UTF-8 and ISO encodings?

A: UTF-8 supports all languages, ISO supports limited Western chars.

Q: Difference between , , and <dl> lists?

A: is ordered, unordered, <dl> description list.

Q: What is progressive enhancement vs graceful degradation?

A: Enhancement builds simple first, degradation builds rich first then fallback.

Q: Why is accessibility (A11y) important in HTML?

A: Helps disabled users (screen readers, keyboards) and improves SEO.

Section 2: CSS (25 Questions)

Q: What is CSS and why is it important?

A: CSS styles HTML pages, separating content from design for maintainability.

Q: Difference between inline, internal, and external CSS?

A: Inline is in tag, internal in <style>, external in .css file linked.

Q: Difference between relative, absolute, fixed, and sticky positioning?

A: Relative shifts from itself, absolute from parent, fixed from viewport, sticky mixes both.

Q: What are pseudo-classes and pseudo-elements?

A: Pseudo-classes select states (:hover), pseudo-elements style parts (::before).

Q: Explain the CSS box model.

A: Element box = content + padding + border + margin.

Q: Difference between inline, block, and inline-block elements?

A: Inline doesn't break line, block takes full width, inline-block mixes both.

Q: Difference between relative units and absolute units?

A: px = fixed, em/rem/% = scalable and responsive.

Q: Difference between z-index and stacking context?

A: z-index works inside same stacking context, created by opacity/position etc.

Q: Difference between relative and absolute paths in CSS?

A: Relative depends on CSS location, absolute is full URL.

Q: Explain difference between CSS Grid and Flexbox.

A: Flexbox handles 1D layouts, Grid handles 2D rows and columns.

Q: What are CSS transitions?

A: Transitions animate property changes smoothly on events.

Q: What are CSS animations?

A: Animations define keyframes for multi-step changes without JS.

Q: Difference between absolute, relative, and fixed units in media queries?

A: Absolute fixed, relative scale with parent/root sizes.

Q: Difference between min-width and max-width in media queries?

A: min-width applies above threshold, max-width below threshold.

Q: Difference between relative and absolute imports in CSS @import?

A: Relative is local path, absolute uses full URL.

Q: What is a CSS reset and why is it used?

A: Removes browser defaults for consistency.

Q: Difference between opacity and visibility:hidden?

A: opacity:0 hides visually but clickable, visibility hides but keeps space.

Q: What are media queries?

A: Rules applied based on screen/device size.

Q: Explain difference between relative and fixed positioning with an example.

A: Relative shifts from normal, fixed locks to viewport (sticky navbars).

Q: Difference between inline styles and CSS classes?

A: Inline has higher priority but not reusable, classes are reusable.

Q: What are CSS variables?

A: Custom reusable values using var(--name).

Q: Difference between relative and absolute font sizing?

A: px = fixed, em/rem/% = scalable and accessible.

Q: Difference between relative vs fixed background images?

A: scroll moves with page, fixed stays in place.

Q: Difference between inline-style, !important, and specificity?

A: Inline > !important > specificity cascade.

Q: Difference between responsive and adaptive design?

A: Responsive uses fluid grids, adaptive uses fixed layouts per device.

Section 3: JavaScript (50 Questions)

Q: What is JavaScript and why is it important?

A: JS makes web pages dynamic, enabling DOM manipulation, async ops, and interactivity.

Q: Difference between var, let, and const?

A: var is function-scoped, let/const block-scoped; const immutable reference.

Q: What are primitive and non-primitive data types in JS?

A: Primitive: string, number, boolean, null, undefined, symbol, bigint. Non-primitive: objects, arrays.

Q: What is the difference between == and ===?

A: == compares with type coercion, === strict without coercion.

Q: What is hoisting in JavaScript?

A: Hoisting moves declarations up; var hoisted undefined, let/const temporal dead zone.

Q: Explain scope in JavaScript.

A: Scope controls access: global, function, block. ES6 adds block scope with let/const.

Q: What is difference between function declaration and expression?

A: Declaration is hoisted, expression not.

Q: What are arrow functions and how do they differ?

A: Concise syntax, no own this, suited for callbacks.

Q: What is the difference between null and undefined?

A: null = intentional empty, undefined = declared but not assigned.

Q: What is pass by value vs pass by reference?

A: Primitives pass by value, objects pass by reference.

Q: Difference between synchronous and asynchronous JS?

A: Sync executes line by line, async uses callbacks/promises/await.

Q: What are template literals in ES6?

A: Strings with backticks supporting interpolation and multi-line.

Q: Difference between map, forEach, and filter?

A: map returns new array, forEach executes only, filter returns matching items.

Q: Difference between slice and splice?

A: slice copies, splice mutates array.

Q: Difference between call, apply, and bind?

A: call passes args list, apply array, bind returns bound function.

Q: What are closures?

A: Function remembering outer scope variables after scope ends.

Q: Difference between this in normal vs arrow function?

A: Normal this = caller, arrow this = lexical context.

Q: What are default parameters?

A: Parameters with fallback values if not passed.

Q: Difference between shallow copy and deep copy?

A: Shallow copies only top level, deep copies nested structures.

Q: What is event loop?

A: Manages async tasks by checking call stack and queues.

Q: Difference between event bubbling and capturing?

A: Bubbling: target->root, Capturing: root->target.

Q: Difference between innerHTML, innerText, and textContent?

A: innerHTML includes HTML, innerText respects style, textContent raw text.

Q: Difference between setTimeout and setInterval?

A: setTimeout runs once, setInterval repeats until cleared.

Q: What are ES6 modules?

A: Uses export/import to modularize code.

Q: Difference between sync vs async callbacks?

A: Sync executes immediately, async later via queue.

Q: What are Promises in JavaScript?

A: Represents async result with states: pending, fulfilled, rejected.

Q: Difference between callback and promise?

A: Callbacks lead to nesting, promises allow chaining then/catch.

Q: What is async/await?

A: Syntax sugar over promises, makes async look sync.

Q: Difference between microtask and macrotask queue?

A: Microtask (promises) run before macrotask (setTimeout).

Q: Difference between undefined and not defined?

A: undefined = declared but not assigned, not defined = undeclared.

Q: Explain prototype in JS.

A: Objects inherit properties via prototype chain.

Q: Difference between __proto__ and prototype?

A: __proto__ is link, prototype is constructor property.

Q: What is prototypal inheritance?

A: Objects inherit directly from other objects.

Q: Difference between classical and prototypal OOP?

A: Classical uses classes, JS uses prototype chaining.

Q: Difference between shallow comparison and deep comparison?

A: Shallow checks reference, deep checks nested equality.

Q: Difference between for..in, for..of, and forEach.

A: for..in keys, for..of values, forEach array iteration.

Q: What is memoization?

A: Caching function outputs for performance.

Q: Difference between pure and impure functions?

A: Pure: same output, no side effects. Impure: depends on external state.

Q: Difference between Object.freeze and Object.seal?

A: freeze immutable, seal modifiable props but no add/remove.

Q: What is event delegation?

A: Using parent listener to handle child events.

Q: What are WeakMap and WeakSet?

A: Store weak references for GC, unlike Map/Set.

Q: Difference between debounce and throttle?

A: Debounce waits after last call, throttle limits execution rate.

Q: Difference between sync vs async iterators?

A: Sync uses next(), async uses await next().

Q: Difference between fetch and axios?

A: fetch native but manual JSON, axios has parsing, interceptors.

Q: Difference between deep and shallow clone?

A: Shallow only top level, deep copies nested fully.

Q: Output: []+[], []+{}, {}+[]

A: "", "[object Object]", "[object Object]".

Q: Output: typeof NaN, NaN===NaN

A: typeof NaN="number", NaN===NaN=false.

Q: Output: 0.1+0.2===0.3

A: false due to floating-point precision issues.

Q: Output: modifying object via reference

A: Both vars point to same object, changes reflected.

Q: What are JS best practices?

A: Use let/const, modularize code, avoid globals, handle errors, use async/await.